SKIM: Efficient Video QA via Reinforcement Learning

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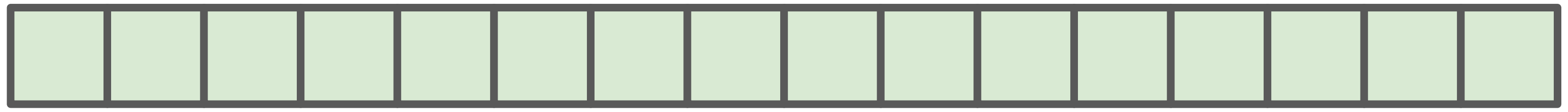
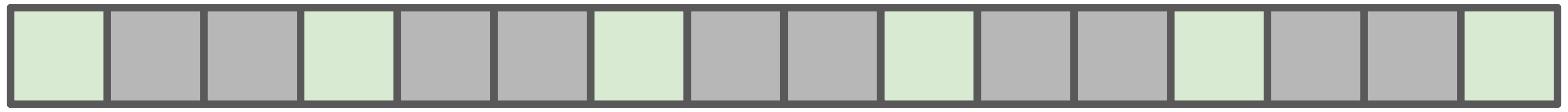
Background on Video QA

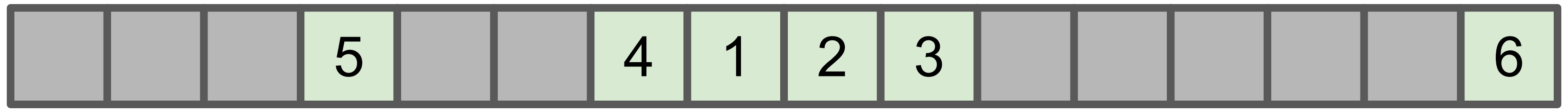
Passing all frames = inefficient →



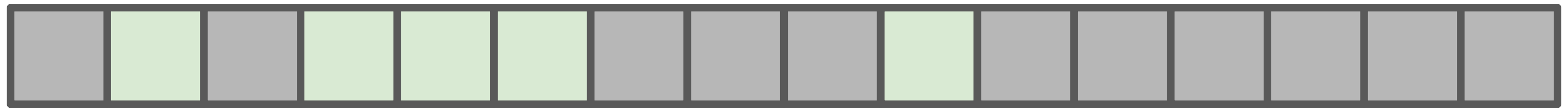
- **Method 1:** Pass frames uniformly (lower frame-rate)



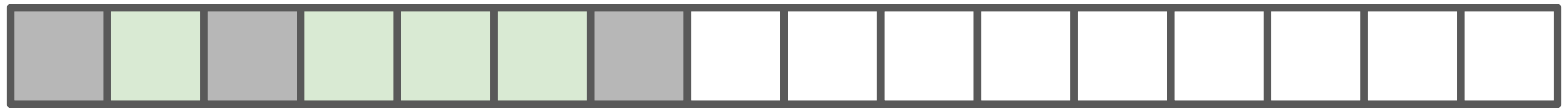
- **Method 2:** Text-frame matching → K frames with embeddings closest to query



- **Method 3:** Model loops through all frames, selects variable-sized subset

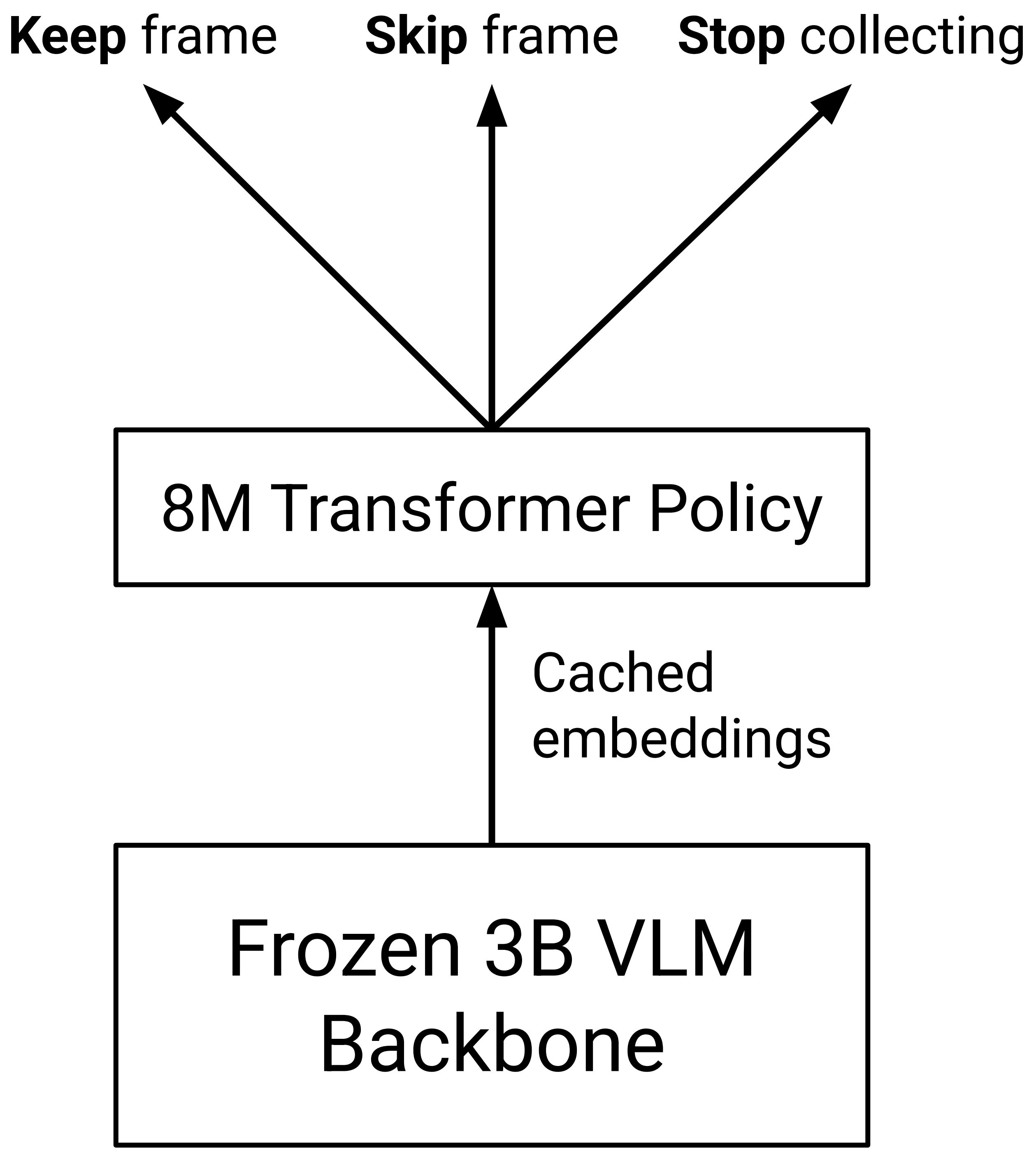


- **Our Goal:** Lightweight model to loop through frames sequentially, potentially stopping



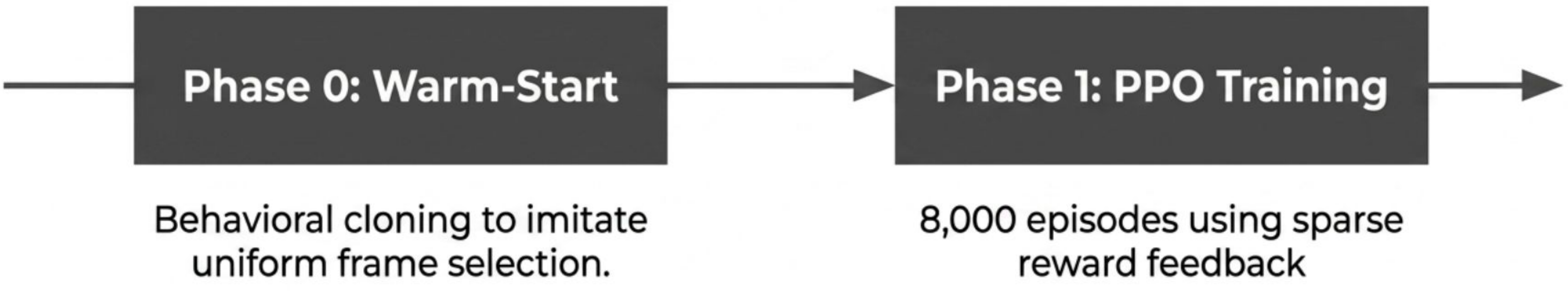
System Architecture

- **Reward:** VLM called at the end of each episode for final answer, +1 for correct answer
  - Penalty: fraction of frames kept * **frame cost (λ)**
- **Selector Model:** 2-Layer Transformer with 8M parameters
  - <u>Input</u> → query + embedded frame + selected frames + progress scalars
  - <u>Output</u> → 3 layer softmax action head
- **Backbone:** Frozen Qwen2.5-VL-3B-Instruct
- **Dataset:** Next-QA for training, evaluate on Next-QA, IntentQA, and EgoSchema (in progress)



Training Progression

- **Warm start** to imitate uniform selection and provide the policy with a baseline
- **8000 episodes of training**, KL divergence to limit divergence from warm start
  - Prevents catastrophic forgetting
- Standard PPO-clip using Generalized Advantage Estimation (GAE)



Results

NExT-QA (n=200)

Method	Accuracy	Avg Frames	Causal	Temporal	Descriptive
Uniform-8	75.0%	8.0	75.0%	69.4%	90.5%
Uniform-16	76.0%	15.7	76.0%	70.8%	89.3%
Uniform-32	78.0%	27.4	77.0%	75.0%	89.3%
PPO λ=0.2	58.5%	1.5	57.0%	54.2%	75.0%
PPO λ=0.1	73.5%	7.1	74.0%	68.1%	85.7%
PPO λ=0.05	72.5%	10.2	76.0%	68.1%	71.4%

Results

IntentQA (n=200)

Method	Accuracy	Avg Frames	Causal	Temporal
Uniform-8	87.5%	8.0	90.1%	79.2%
Uniform-16	88.0%	15.8	89.5%	83.3%
Uniform-32	90.0%	29.2	90.8%	87.5%
PPO λ=0.2	78.5%	2.0	84.2%	60.4%
PPO λ=0.1	88.5%	11.1	89.5%	85.4%
PPO λ=0.05	90.0%	17.2	90.8%	87.5%

Analyzing Early Stoppage

- Too high of a frame cost results in the policy stopping after taking 0-1 frames
- Causal questions remain relatively unchanged
- Temporal questions hurt because they need to see frames from the entire video
  - Accuracy is higher on IntentQA because the policy stops 20% less than NExT-QA
- **Descriptive questions best show the potential of early stoppage**

PPO λ=0.05

Metrics	**Average Frames**	**Descriptive Accuracy**
STOP Action Fires	7.1	93.3%
No STOP Action	13.0	46.2%

Conclusion

- Reward hacking is a problem with stopping early, the policy can stop after selecting 0 frames and use the VLM’s language prior
- The descriptive questions using PPO λ=0.05 shows that early stopping can work in principle on these types of questions
- Learning how to more reliably distinguish between when to stop early and when to ingest the full context is important

Remaining Steps

- Evaluate on full validation sets of NExT-QA/IntentQA (not the n=200 subsets)
- Evaluate on EgoSchema

Thank you!